

Machine learning-based analysis of calcium sulfate scaling-induced permeability decline in semi-batch reverse osmosis

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S1. LODO evaluation with temporal variables

Table S1. 5-fold cross-validation and LODO evaluation performance of the M5/2GPR model with temporal variables.

Held-out experimental dataset	5-fold cross-validation			LODO evaluation		
	R ² [–]	RMSE [–]	MAE [–]	R ² [–]	RMSE [–]	MAE [–]
1	0.998	0.003	0.002	0.963	0.014	0.011
2	0.999	0.003	0.002	0.831	0.025	0.023
3	0.999	0.003	0.002	–0.969	0.081	0.078
4	0.998	0.003	0.002	0.292	0.078	0.059
5	0.998	0.003	0.002	–0.125	0.060	0.052
6	0.998	0.003	0.002	0.317	0.048	0.046
7	0.998	0.003	0.002	0.795	0.028	0.027
8	0.998	0.003	0.002	0.551	0.042	0.035
9	0.998	0.003	0.002	0.935	0.016	0.014
10	0.998	0.003	0.002	0.818	0.017	0.014
11	0.998	0.003	0.002	0.439	0.037	0.035
12	0.998	0.003	0.002	0.999	0.002	0.002
Mean ± SD	0.998 ± 0.000	0.003 ± 0.000	0.002 ± 0.000	0.487 ± 0.571	0.037 ± 0.025	0.033 ± 0.023

Table S2. 5-fold cross-validation and LODO evaluation performance of the RQGPR model with temporal variables.

Held-out experimental dataset	5-fold cross-validation			LODO evaluation		
	R ² [–]	RMSE [–]	MAE [–]	R ² [–]	RMSE [–]	MAE [–]
1	0.998	0.003	0.002	0.950	0.016	0.013
2	0.999	0.003	0.002	0.839	0.025	0.023
3	0.998	0.003	0.002	–0.842	0.078	0.075
4	0.998	0.003	0.002	0.356	0.074	0.056
5	0.998	0.003	0.002	–0.203	0.062	0.053
6	0.998	0.003	0.002	0.295	0.049	0.046
7	0.998	0.003	0.002	0.833	0.026	0.024
8	0.998	0.003	0.002	0.579	0.041	0.032
9	0.998	0.003	0.002	0.921	0.018	0.016
10	0.998	0.003	0.002	0.835	0.017	0.013
11	0.998	0.003	0.002	0.471	0.036	0.034
12	0.998	0.003	0.002	0.999	0.002	0.002
Mean ± SD	0.998 ± 0.000	0.003 ± 0.000	0.002 ± 0.000	0.503 ± 0.550	0.037 ± 0.024	0.032 ± 0.022

Table S3. 5-fold cross-validation and LODO evaluation performance of the MNN model with temporal variables.

Held-out experimental dataset	5-fold cross-validation			LODO evaluation		
	R ² [–]	RMSE [–]	MAE [–]	R ² [–]	RMSE [–]	MAE [–]
1	0.996	0.004	0.003	0.722	0.038	0.034
2	0.996	0.004	0.003	0.838	0.025	0.021
3	0.996	0.004	0.003	–0.103	0.060	0.058
4	0.995	0.005	0.003	0.173	0.084	0.071
5	0.995	0.005	0.003	–0.458	0.068	0.064
6	0.995	0.005	0.004	0.457	0.043	0.035
7	0.995	0.005	0.004	–0.145	0.067	0.051
8	0.996	0.004	0.003	0.649	0.037	0.027
9	0.996	0.004	0.003	0.829	0.026	0.018
10	0.995	0.005	0.004	0.738	0.021	0.017
11	0.995	0.005	0.004	0.083	0.047	0.047
12	0.994	0.005	0.004	0.996	0.004	0.003
Mean ± SD	0.995 ±	0.005 ±	0.003 ±	0.398 ±	0.043 ±	0.037 ±
	0.001	0.000	0.000	0.472	0.023	0.021

Table S4. 5-fold cross-validation and LODO evaluation performance of the fine tree model with temporal variables.

Held-out experimental dataset	5-fold cross-validation			LODO evaluation		
	R ² [–]	RMSE [–]	MAE [–]	R ² [–]	RMSE [–]	MAE [–]
1	0.992	0.006	0.004	0.820	0.030	0.024
2	0.992	0.006	0.004	0.862	0.023	0.018
3	0.992	0.006	0.004	0.495	0.041	0.035
4	0.993	0.005	0.003	0.267	0.079	0.064
5	0.992	0.006	0.004	–0.494	0.069	0.059
6	0.990	0.007	0.004	0.106	0.055	0.049
7	0.990	0.007	0.004	0.724	0.033	0.028
8	0.990	0.007	0.004	0.534	0.043	0.038
9	0.991	0.007	0.004	0.790	0.029	0.023
10	0.991	0.006	0.004	0.764	0.020	0.017
11	0.991	0.007	0.004	0.066	0.048	0.045
12	0.988	0.007	0.005	0.991	0.006	0.004
Mean ± SD	0.991 ±	0.006 ±	0.004 ±	0.494 ±	0.040 ±	0.034 ±
	0.001	0.000	0.000	0.433	0.021	0.018

Table S5. 5-fold cross-validation and LODO evaluation performance of the MGSVM model with temporal variables.

Held-out experimental dataset	5-fold cross-validation			LODO evaluation		
	R ² [–]	RMSE [–]	MAE [–]	R ² [–]	RMSE [–]	MAE [–]
1	0.990	0.007	0.005	0.819	0.030	0.026
2	0.987	0.008	0.005	0.410	0.047	0.047
3	0.988	0.007	0.005	0.443	0.043	0.041
4	0.991	0.006	0.004	0.069	0.089	0.071
5	0.991	0.006	0.005	–0.477	0.069	0.060
6	0.986	0.008	0.005	0.512	0.041	0.039
7	0.989	0.007	0.005	0.620	0.039	0.029
8	0.992	0.006	0.004	0.297	0.053	0.047
9	0.986	0.008	0.005	0.847	0.025	0.022
10	0.987	0.008	0.005	0.918	0.012	0.011
11	0.988	0.007	0.005	0.280	0.042	0.041
12	0.988	0.007	0.004	0.997	0.004	0.002
Mean ± SD	0.989 ±	0.007 ±	0.005 ±	0.478 ±	0.041 ±	0.036 ±
	0.002	0.001	0.000	0.415	0.023	0.020

Table S6. 5-fold cross-validation and LODO evaluation performance of the SVM kernel model with temporal variables.

Held-out experimental dataset	5-fold cross-validation			LODO evaluation		
	R ² [–]	RMSE [–]	MAE [–]	R ² [–]	RMSE [–]	MAE [–]
1	0.975	0.010	0.007	–0.864	0.098	0.079
2	0.978	0.010	0.007	–1.419	0.096	0.076
3	0.984	0.009	0.006	–3.258	0.119	0.100
4	0.984	0.008	0.006	–0.620	0.118	0.099
5	0.988	0.007	0.006	–2.640	0.108	0.083
6	0.984	0.008	0.006	–1.426	0.090	0.075
7	0.985	0.008	0.006	–0.093	0.066	0.054
8	0.973	0.011	0.008	–1.228	0.094	0.070
9	0.986	0.008	0.006	0.161	0.058	0.043
10	0.980	0.010	0.007	–7.370	0.119	0.101
11	0.979	0.010	0.007	0.158	0.046	0.038
12	0.977	0.010	0.007	0.990	0.007	0.005
Mean ± SD	0.981 ±	0.009 ±	0.007 ±	–1.468 ±	0.085 ±	0.069 ±
	0.005	0.001	0.001	2.211	0.034	0.029

Table S7. 5-fold cross-validation and LODO evaluation performance of the QSVM model with temporal variables.

Held-out experimental dataset	5-fold cross-validation			LODO evaluation		
	R ² [–]	RMSE [–]	MAE [–]	R ² [–]	RMSE [–]	MAE [–]
1	0.960	0.013	0.010	–0.152	0.077	0.066
2	0.915	0.020	0.015	0.347	0.050	0.046
3	0.951	0.015	0.011	–1.593	0.093	0.092
4	0.926	0.017	0.013	–7.386	0.268	0.261
5	0.974	0.011	0.008	–1.533	0.090	0.089
6	0.904	0.021	0.015	0.600	0.037	0.025
7	0.881	0.023	0.017	0.366	0.050	0.043
8	0.966	0.012	0.010	0.294	0.053	0.050
9	0.934	0.017	0.013	0.417	0.048	0.046
10	0.916	0.020	0.015	0.828	0.017	0.013
11	0.914	0.020	0.015	–0.719	0.065	0.061
12	0.945	0.016	0.012	0.893	0.022	0.018
Mean ± SD	0.932 ±	0.017 ±	0.013 ±	–0.636 ±	0.072 ±	0.067 ±
	0.028	0.004	0.003	2.286	0.066	0.066

Table S8. 5-fold cross-validation and LODO evaluation performance of the linear model with temporal variables.

Held-out experimental dataset	5-fold cross-validation			LODO evaluation		
	R ² [–]	RMSE [–]	MAE [–]	R ² [–]	RMSE [–]	MAE [–]
1	0.872	0.024	0.018	0.719	0.038	0.036
2	0.869	0.024	0.019	0.667	0.035	0.030
3	0.882	0.023	0.018	0.618	0.036	0.034
4	0.887	0.021	0.017	0.363	0.074	0.059
5	0.875	0.024	0.019	0.774	0.027	0.024
6	0.904	0.021	0.015	–0.027	0.059	0.058
7	0.889	0.022	0.018	0.432	0.047	0.042
8	0.882	0.023	0.018	0.712	0.034	0.033
9	0.863	0.025	0.020	0.975	0.010	0.008
10	0.873	0.024	0.019	0.847	0.016	0.014
11	0.907	0.021	0.016	–0.553	0.062	0.062
12	0.874	0.024	0.019	0.875	0.024	0.019
Mean ± SD	0.881 ±	0.023 ±	0.018 ±	0.534 ±	0.038 ±	0.035 ±
	0.014	0.002	0.001	0.436	0.019	0.018

Table S9. Summary of 5-fold cross-validation and LODO evaluation performance of the regression ML models with temporal variables.

ML models	5-fold cross-validation			LODO evaluation		
	R^2 [-]	RMSE [-]	MAE [-]	R^2 [-]	RMSE [-]	MAE [-]
M5/2GPR	0.998 ± 0.000	0.003 ± 0.000	0.002 ± 0.000	0.487 ± 0.571	0.037 ± 0.025	0.033 ± 0.023
RQGPR	0.998 ± 0.000	0.003 ± 0.000	0.002 ± 0.000	0.503 ± 0.550	0.037 ± 0.024	0.032 ± 0.022
MNN	0.995 ± 0.001	0.005 ± 0.000	0.004 ± 0.000	0.398 ± 0.472	0.043 ± 0.023	0.037 ± 0.021
Fine tree	0.991 ± 0.001	0.006 ± 0.001	0.004 ± 0.000	0.494 ± 0.433	0.040 ± 0.021	0.034 ± 0.018
MG SVM	0.989 ± 0.002	0.007 ± 0.001	0.005 ± 0.000	0.478 ± 0.415	0.041 ± 0.023	0.036 ± 0.020
SVM kernel	0.981 ± 0.005	0.009 ± 0.001	0.007 ± 0.001	-1.468 ± 2.211	0.085 ± 0.034	0.069 ± 0.029
QSVM	0.932 ± 0.028	0.017 ± 0.004	0.013 ± 0.003	-0.636 ± 2.286	0.072 ± 0.066	0.067 ± 0.066
Linear	0.881 ± 0.014	0.023 ± 0.002	0.018 ± 0.002	0.534 ± 0.436	0.038 ± 0.019	0.035 ± 0.018

Table S10. Performance of the regression ML models obtained by 5-fold cross-validation using the complete set of 12 experiments with temporal variables.

ML models	5-fold cross-validation		
	R^2 [-]	RMSE [-]	MAE [-]
M5/2GPR	0.998	0.003	0.002
RQGPR	0.998	0.003	0.002
MNN	0.995	0.005	0.004
Fine tree	0.990	0.007	0.004
MG SVM	0.987	0.008	0.005
SVM kernel	0.977	0.010	0.007
QSVM	0.935	0.017	0.013
Linear	0.875	0.024	0.019

S2. LODO evaluation without temporal variables

Table S11. 5-fold cross-validation and LODO evaluation performance of the M5/2GPR model without temporal variables.

Held-out experimental dataset	5-fold cross-validation			LODO evaluation		
	R ² [–]	RMSE [–]	MAE [–]	R ² [–]	RMSE [–]	MAE [–]
1	0.998	0.003	0.002	0.901	0.023	0.019
2	0.998	0.003	0.002	0.861	0.023	0.021
3	0.998	0.003	0.002	–0.371	0.067	0.066
4	0.998	0.003	0.002	0.303	0.077	0.059
5	0.998	0.003	0.002	–0.126	0.060	0.053
6	0.998	0.003	0.002	0.610	0.036	0.029
7	0.998	0.003	0.002	0.811	0.027	0.026
8	0.998	0.003	0.002	0.494	0.045	0.040
9	0.998	0.003	0.002	0.915	0.018	0.016
10	0.998	0.003	0.002	0.797	0.018	0.016
11	0.998	0.003	0.002	0.340	0.040	0.039
12	0.998	0.003	0.002	0.931	0.016	0.013
Mean ± SD	0.998 ± 0.000	0.003 ± 0.000	0.002 ± 0.000	0.539 ± 0.431	0.038 ± 0.021	0.033 ± 0.018

Table S12. 5-fold cross-validation and LODO evaluation performance of the RQGPR model without temporal variables.

Held-out experimental dataset	5-fold cross-validation			LODO evaluation		
	R ² [–]	RMSE [–]	MAE [–]	R ² [–]	RMSE [–]	MAE [–]
1	0.998	0.003	0.002	0.903	0.022	0.019
2	0.998	0.003	0.002	0.843	0.024	0.022
3	0.998	0.003	0.002	–0.349	0.067	0.065
4	0.998	0.003	0.002	0.340	0.075	0.057
5	0.998	0.003	0.002	–0.251	0.063	0.055
6	0.998	0.003	0.002	0.576	0.038	0.029
7	0.998	0.003	0.002	0.849	0.024	0.022
8	0.998	0.003	0.002	0.518	0.044	0.038
9	0.998	0.003	0.002	0.896	0.020	0.017
10	0.998	0.003	0.002	0.791	0.019	0.016
11	0.998	0.003	0.002	0.394	0.039	0.037
12	0.998	0.003	0.002	0.921	0.017	0.013
Mean ± SD	0.998 ± 0.000	0.003 ± 0.000	0.002 ± 0.000	0.536 ± 0.440	0.038 ± 0.021	0.033 ± 0.018

Table S13. 5-fold cross-validation and LODO evaluation performance of the MNN model without temporal variables.

Held-out experimental dataset	5-fold cross-validation			LODO evaluation		
	R ² [–]	RMSE [–]	MAE [–]	R ² [–]	RMSE [–]	MAE [–]
1	0.996	0.004	0.003	0.680	0.041	0.031
2	0.996	0.005	0.003	–2.441	0.114	0.112
3	0.996	0.004	0.003	–0.910	0.079	0.077
4	0.995	0.004	0.003	0.178	0.084	0.080
5	0.996	0.004	0.003	–1.307	0.086	0.084
6	0.995	0.005	0.003	–0.124	0.062	0.056
7	0.996	0.004	0.003	0.555	0.042	0.031
8	0.996	0.004	0.003	0.664	0.037	0.024
9	0.996	0.005	0.003	0.962	0.012	0.010
10	0.996	0.004	0.003	0.856	0.016	0.014
11	0.996	0.004	0.003	–0.550	0.062	0.061
12	0.995	0.005	0.004	0.630	0.036	0.030
Mean ± SD	0.996 ±	0.004 ±	0.003 ±	–0.067 ±	0.056 ±	0.051 ±
	0.000	0.000	0.000	1.047	0.031	0.032

Table S14. 5-fold cross-validation and LODO evaluation performance of the fine tree model without temporal variables.

Held-out experimental dataset	5-fold cross-validation			LODO evaluation		
	R ² [–]	RMSE [–]	MAE [–]	R ² [–]	RMSE [–]	MAE [–]
1	0.989	0.007	0.004	0.820	0.030	0.023
2	0.990	0.007	0.004	0.674	0.035	0.031
3	0.991	0.007	0.004	0.643	0.034	0.031
4	0.993	0.005	0.003	0.262	0.079	0.064
5	0.992	0.006	0.004	–0.593	0.071	0.064
6	0.990	0.007	0.004	0.512	0.041	0.037
7	0.990	0.007	0.004	0.794	0.029	0.024
8	0.988	0.007	0.004	0.539	0.043	0.037
9	0.991	0.006	0.004	0.853	0.024	0.021
10	0.988	0.007	0.004	0.539	0.028	0.025
11	0.991	0.006	0.004	–0.249	0.055	0.054
12	0.991	0.006	0.004	0.820	0.025	0.021
Mean ± SD	0.990 ±	0.007 ±	0.004 ±	0.468 ±	0.041 ±	0.036 ±
	0.001	0.001	0.000	0.454	0.018	0.016

Table S15. 5-fold cross-validation and LODO evaluation performance of the MGSVM model without temporal variables.

Held-out experimental dataset	5-fold cross-validation			LODO evaluation		
	R ² [–]	RMSE [–]	MAE [–]	R ² [–]	RMSE [–]	MAE [–]
1	0.992	0.006	0.004	0.719	0.038	0.034
2	0.989	0.007	0.005	0.469	0.045	0.044
3	0.991	0.007	0.004	0.319	0.047	0.046
4	0.992	0.006	0.004	0.093	0.088	0.071
5	0.992	0.006	0.004	–0.488	0.069	0.060
6	0.989	0.007	0.005	0.671	0.033	0.032
7	0.991	0.006	0.004	0.600	0.040	0.030
8	0.993	0.006	0.004	0.321	0.052	0.045
9	0.989	0.007	0.005	0.854	0.024	0.022
10	0.990	0.007	0.005	0.936	0.010	0.008
11	0.990	0.007	0.005	0.244	0.043	0.042
12	0.990	0.007	0.005	0.825	0.025	0.024
Mean ± SD	0.991 ±	0.006 ±	0.005 ±	0.464 ±	0.043 ±	0.038 ±
	0.001	0.001	0.000	0.401	0.021	0.017

Table S16. 5-fold cross-validation and LODO evaluation performance of the SVM kernel model without temporal variables.

Held-out experimental dataset	5-fold cross-validation			LODO evaluation		
	R ² [–]	RMSE [–]	MAE [–]	R ² [–]	RMSE [–]	MAE [–]
1	0.981	0.009	0.006	–0.470	0.087	0.063
2	0.977	0.010	0.007	–2.072	0.108	0.096
3	0.987	0.008	0.006	–3.204	0.118	0.098
4	0.976	0.010	0.006	–0.820	0.125	0.099
5	0.981	0.009	0.007	–0.489	0.069	0.057
6	0.978	0.010	0.007	–0.433	0.069	0.058
7	0.984	0.009	0.006	–4.474	0.147	0.112
8	0.977	0.010	0.007	–0.204	0.069	0.060
9	0.984	0.008	0.006	–1.746	0.105	0.088
10	0.985	0.008	0.006	–3.542	0.087	0.079
11	0.988	0.007	0.005	0.591	0.032	0.027
12	0.981	0.009	0.006	–1.142	0.087	0.078
Mean ± SD	0.982 ±	0.009 ±	0.006 ±	–1.500 ±	0.092 ±	0.076 ±
	0.004	0.001	0.000	1.542	0.031	0.024

Table S17. 5-fold cross-validation and LODO evaluation performance of the QSVM model without temporal variables.

Held-out experimental dataset	5-fold cross-validation			LODO evaluation		
	R ² [–]	RMSE [–]	MAE [–]	R ² [–]	RMSE [–]	MAE [–]
1	0.954	0.014	0.010	0.244	0.062	0.058
2	0.892	0.022	0.017	0.861	0.023	0.018
3	0.938	0.017	0.013	–1.546	0.092	0.090
4	0.875	0.022	0.017	–8.720	0.288	0.284
5	0.932	0.017	0.013	–1.827	0.095	0.093
6	0.895	0.022	0.016	0.317	0.048	0.033
7	0.906	0.021	0.016	0.110	0.059	0.053
8	0.953	0.015	0.011	–0.219	0.070	0.068
9	0.892	0.022	0.017	0.403	0.049	0.045
10	0.905	0.021	0.016	0.842	0.016	0.014
11	0.883	0.023	0.018	–0.279	0.056	0.051
12	0.876	0.024	0.018	0.306	0.050	0.040
Mean ± SD	0.908 ±	0.020 ±	0.015 ±	–0.792 ±	0.076 ±	0.071 ±
	0.029	0.003	0.003	2.633	0.071	0.071

Table S18. 5-fold cross-validation and LODO evaluation performance of the linear model without temporal variables.

Held-out experimental dataset	5-fold cross-validation			LODO evaluation		
	R ² [–]	RMSE [–]	MAE [–]	R ² [–]	RMSE [–]	MAE [–]
1	0.870	0.024	0.019	0.747	0.036	0.034
2	0.863	0.025	0.020	0.924	0.017	0.015
3	0.880	0.023	0.018	0.629	0.035	0.033
4	0.886	0.021	0.017	0.378	0.073	0.058
5	0.872	0.024	0.019	0.803	0.025	0.023
6	0.901	0.021	0.016	0.097	0.055	0.054
7	0.888	0.022	0.017	0.432	0.047	0.042
8	0.880	0.023	0.018	0.721	0.033	0.032
9	0.861	0.025	0.020	0.981	0.009	0.007
10	0.871	0.024	0.019	0.912	0.012	0.011
11	0.906	0.021	0.016	–0.540	0.062	0.061
12	0.872	0.024	0.019	0.879	0.021	0.017
Mean ± SD	0.879 ±	0.023 ±	0.018 ±	0.580 ±	0.035 ±	0.032 ±
	0.014	0.002	0.001	0.440	0.020	0.019

Table S19. Performance of the regression ML models obtained by 5-fold cross-validation using the complete set of 12 experiments without temporal variables.

ML models	5-fold cross-validation		
	R^2 [–]	RMSE [–]	MAE [–]
M5/2GPR	0.998	0.003	0.000
RQGPR	0.998	0.003	0.000
MNN	0.995	0.005	0.000
Fine tree	0.991	0.006	0.000
MG SVM	0.990	0.007	0.000
SVM kernel	0.983	0.009	0.000
QSVM	0.910	0.020	0.000
Linear	0.873	0.024	0.001